

Calculation Supplement — Document 5

## **TOPSIS Multicriteria Analysis**

Nine methods, seven criteria, and three weighting schemes

343 runs / 9 methods / 4 scenarios

### **Step-by-step multicriteria analysis**

of social navigation in an agricultural digital twin

**Purpose.** To support the calculations of the  $9 \times 7$  TOPSIS with its criteria, three weight vectors, exact closeness coefficients, and ranks.

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## 1 Analysis configuration

Rows of  $X \in \mathbb{R}^{9 \times 7}$  are M0–M4 and B1–B4. Columns are  $(d^*, T_{personal}, T_{mission}, N_{stop}, \sigma_v, a_{max}^{num}, \eta_s)$ .  $d^*$  and  $\eta_s$  are benefits; all others are costs.

$$w_{W1} = (0.20, 0.15, 0.10, 0.15, 0.15, 0.10, 0.15), \quad (1)$$

$$w_{W2} = (0.22, 0.20, 0.05, 0.10, 0.20, 0.15, 0.08), \quad (2)$$

$$w_{W3} = (0.20, 0.15, 0.05, 0.10, 0.25, 0.20, 0.05). \quad (3)$$

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\sum_k x_{kj}^2}}, \quad v_{ij} = w_j \tilde{x}_{ij}, \quad CC_i = \frac{D_i^-}{D_i^+ + D_i^-}. \quad (4)$$

## 2 Complete procedure

1. Build  $X$  from success-conditional means and success rate  $\eta_s$ .
2. Mark  $(d^*, \eta_s)$  as benefits and all remaining criteria as costs.
3. Normalize each column by its Euclidean norm.
4. Multiply each column by weight  $w_j$ .
5. Build  $A^+$  from benefit maxima and cost minima.
6. Build  $A^-$  from benefit minima and cost maxima.
7. Compute  $D_i^+ = \sqrt{\sum_j (v_{ij} - A_j^+)^2}$ .
8. Compute  $D_i^- = \sqrt{\sum_j (v_{ij} - A_j^-)^2}$ .
9. Obtain  $CC_i$  and sort from highest to lowest.

The calculation is repeated for W1, W2, and W3 without altering  $X$ , so rank changes arise only from the represented priorities.

## 3 Results

**Table 1:** Closeness coefficient and rank.

| Method | W1 primary |      | W2 social |      | W3 kinematic |      |
|--------|------------|------|-----------|------|--------------|------|
|        | CC         | Rank | CC        | Rank | CC           | Rank |
| B4     | 0.781      | 1    | 0.733     | 1    | 0.658        | 3    |
| B3     | 0.767      | 2    | 0.727     | 2    | 0.727        | 1    |
| M3     | 0.751      | 3    | 0.668     | 4    | 0.616        | 4    |
| B1     | 0.730      | 4    | 0.718     | 3    | 0.689        | 2    |
| M4     | 0.711      | 5    | 0.619     | 5    | 0.599        | 5    |
| M0     | 0.595      | 6    | 0.492     | 6    | 0.543        | 6    |
| B2     | 0.551      | 7    | 0.448     | 7    | 0.509        | 7    |
| M2     | 0.514      | 8    | 0.436     | 8    | 0.375        | 8    |
| M1     | 0.213      | 9    | 0.248     | 9    | 0.218        | 9    |

## 4 Interpretation

W1 ranks B4 first, B3 second, M3 third, B1 fourth, and M4 fifth. W3 rewards kinematic throughput and B3 leads. In the six-method subset, the order is  $(B1, M3, M4, M0, M2, M1)$ . All nine methods are Pareto-optimal in seven-criterion space: leadership expresses priorities, not intrinsic dominance.

## 5 Acceleration caveat

$a_{max}^{num}$  is a finite-difference pseudo-acceleration with unverified cross-source comparability. The analysis treats  $\sigma_v$  as primary smoothness and  $a_{max}^{num}$  as secondary.

## 6 Conclusion

Ranking is weight-dependent. B4/B3 lead under W1/W2; B3 leads W3; M4 occupies a short-path, perfectly reliable, intermediate-safety compromise.

## 7 Reproducibility control

Weights must sum to one, benefit/cost directions must not be inverted, and normalized values must not be rounded before distance calculations. Reproduction should recover every displayed  $CC$  within numerical tolerance and preserve rank ordering.